# When Competition Compels Change: Trade, Management, and Productivity

Ananya Kotia
Stanford University

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Princeton Development Seminar

# Gains from Trade “Inside the Firm”

*“The idea that* ***international trade increases competition*** *goes back to Adam Smith, and is one of the reasons why economists believe that the* ***gains from trade and the costs from protection are larger than what their own models suggest****.”* —Helpman and Krugman (1989)

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- Selection: competition drives out laggard firms (Melitz, 2003)

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- Selection: competition drives out laggard firms (Melitz, 2003)
- Within-firm "Slack" or "X-inefficiency"
  - *"Absent competitive pressure, firms may not use their resources efficiently"* (Leibenstein, 1966)
- **This paper:** competition compels firms to improve management quality and firm organization
  ⟶ accompanied by within-firm productivity growth
- In today's global context, understanding how trade shapes organizations is increasingly urgent

# Management Quality and Firm Organization

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**Shri George Jacob Muthoot**

Group Chairman
The Muthoot Group of Companies



V.A. George
Independent Director
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George Alexander
Deputy Managing Director
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George Muthoot Jacob
Deputy Managing Director
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George Muthoot George
Deputy Managing Director
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Alexander George Muthoot
Joint Managing Director
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George Thomas Muthoot
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George Alexander Muthoot
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# Management Quality and Firm Organization



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# Management Quality and Firm Organization: The House of Georges

- Why do firms organize this way?
  - Owners derive non-monetary private benefits from family management (Demsetz and Lehn, 1985)
- Trade and competition can be a powerful force that motivates such firms to professionalize
  - Competition eliminates the luxury of prioritizing non-monetary private benefits over monetary gains from firm efficiency
  - These organizational changes occur without deeper cultural or institutional shifts

# Foreign Competition Induces Family Firms to Professionalize

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2. **Natural Experiment:** large, product-specific import competition shock → event-study design

3. **Quantitative Framework:** How much does the within-firm response matter in the aggregate?

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   b. Family firms professionalize management: 15% ↓ share of family executives on the board
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3. **Quantitative Framework:** How much does the within-firm response matter in the aggregate?
   - Competition compels low productivity firms to professionalize to avoid exit
   - Impact on productivity and welfare is ambiguous in theory:
     a. Professionalizing firms become more productive
     b. Saving laggards weakens selection
     c. Saving laggards preserves domestic variety

# Contribution to the Literature

- **Competition and Firm Productivity:** Hicks (1935); Leibenstein (1966); Nalebuff and Stiglitz (1983); Hart (1983); Schmidt (1997); Raith (2003); Nickell (1996); Schmitz (2005); Holmes & Schmitz (2010); Backus (2020)

→ Open the black box of *"X-inefficiency"*: managerial upgrading

- **Competition and Managerial Practices:** Bloom and Van Reenen (2007); Bloom et al. (2012, 2013), Abebe et al. (2025)

→ Why do poor managerial practices persist despite high returns? Can competition lead to better management?

- **Pro-competitive Gains From Trade:**
  Across-firm resource reallocation: Melitz (2003); ACR (2012); Edmond et al. (2014); Arkolakis et al. (2019)
  Within-firm productivity gains: Caliendo and Rossi-Hansberg (2012); Caliendo et al. (2015, 2020); Bustos (2011); Akcigit Melitz (2022); Bloom et al. (2016); Topalova Khandelwal (2011); Amiti and Konings (2007); Bernard et al. (2011); Dhingra (2013) Atkin Khandelwal (2017); Alfaro-Urena et al. (2022) Autor et al. (2020); Chen Steinwender (2021)

→ Organizational channel (talent of managers) of within-firm productivity gains from trade

- **Family Firms:** Bertrand Schoar (2006); Burkart et al. (2003); Bennedsen et al. (2007); Pérez-González (2006); Villalonga Amit (2007), Bertrand et al. (2008); Caselli Gennaioli (2013); Bandiera et al. (2018); Lemos Scur (2018); Akcigit et al. (2021)

→ Response of family firms to globalization

# Outline

1. Introduction
2. Natural Experiment
3. Data
4. Empirical Results
5. Model
6. Calibration & Counterfactuals

# Outline

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# Removal of Quantitative Restrictions (QRs): Large, Product-Specific Trade Shock

**1950s-** India used **Quantitative Restrictions (QRs)** to protect domestic firms

**1990s:** First-generation reforms in 1991 removed some QRs; many remained

- 3000 HS 8-digit products, almost 30% of all tariff lines
- Mostly final consumer goods
- QRs continued by exploiting GATT Article XVIII:B (QRs on weak BoP grounds)

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**1997: Second-generation WTO-imposed reforms: removal of all remaining QRs**

- After complaints to WTO by the US and EU

**2000s: Imports of liberalized products soared after the reform. No impact on exports.**

Indian Response | HS Product Code Example | Media Coverage | Empirical Strategy | Timeline and Scope | Impact on Trade

# Removal of quantitative restrictions on all imports to transform Indian market

*Removal of quantitative restrictions on all imports will transform the Indian market like never before.*



PROFUSION OF CHOICE: Shelves like these in a Chennai super store are getting stacked with an array of foreign products

*Amma Naana Super Store, TTK Road, Chennai: Heinz spaghetti with sausages in tomato sauce, 220 g for Rs 155; H-Vollmilch low-fat milk with one-year shelf life,1 litre for Rs 66; Sugar-free chocolate biscuits, 184 g for Rs 98...*

*Rustom Stores, Colaba, Mumbai: Blue Bunnyice cream, 1.5 litre for Rs 425; St Martin's iced tea, 240 ml for Rs 25; Langnese honey, 500 g for Rs 165*

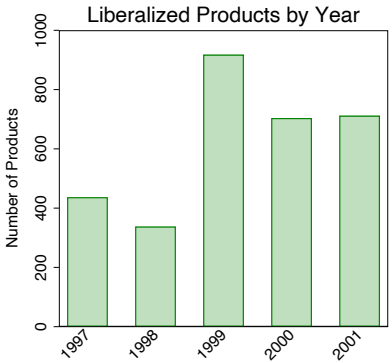
*Steak House, Jor Bagh, Delhi: Lake land mayonnaise, 450 g for Rs 98; Laughing Cow cheese, 180 g for Rs 45...*

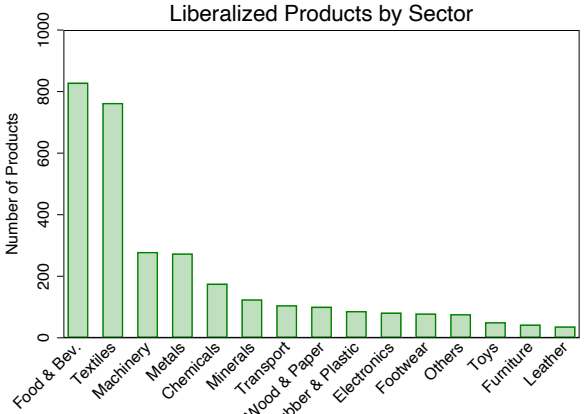
A sample of goodies on the shelves of stores in the three metros on April 3, two days after India abolished all quantitative restrictions (QRs) on imports. On payment of customs duty, anybody can now import any amount of any product. A freedom of economic choice that comes a full 54 years after the freedom of political choice the country attained in 1947.

*Source:* India Today Magazine, April 16, 2001.

Back

# QR-Removal: Timeline and Sectoral Scope




Back

# Unusually Narrow Scope of India's Second-Generation Trade Reforms

- Variation in import competition across the product space and time
- Externally imposed: hinged on IMF technical assessment of India's BoP
- Unlike first-gen reforms, no concurrent domestic deregulation → isolate trade reform impact
- No change in export policy → isolate import competition impact
- Affected mostly consumer goods: limited impact on firms' input costs

HS Product Code Example

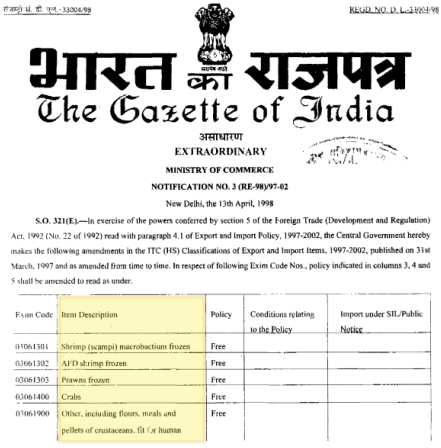
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# Data Construction

1. QR removal: Identify affected products
   - More precise treatment assignment
   - ∼ 200 industries in past work vs. 10k+ products

**Digitized > 1,000 pages of Customs Notifications**



रजिस्ट्री सं. डी. एल.-33004/98

REGD. NO. D. L.-33004/98

भारत का राजपत्र

The Gazette of India

असाधारण

EXTRAORDINARY

MINISTRY OF COMMERCE

NOTIFICATION NO. 3 (RE-98)/97-02

New Delhi, the 13th April, 1998

S.O. 321(E).—In exercise of the powers conferred by section 5 of the Foreign Trade (Development and Regulation) Act, 1992 (No. 22 of 1992) read with paragraph 4.1 of Export and Import Policy, 1997-2002, the Central Government hereby makes the following amendments in the ITC (HS) Classifications of Export and Import Items, 1997-2002, published on 31st March, 1997 and as amended from time to time. In respect of following Exim Code Nos., policy indicated in columns 3, 4 and 5 shall be amended to read as under:

| Exim Code | Item Description | Policy | Conditions relating to the Policy | Import under SIL/Public Notice |
|---|---|---|---|---|
| 03061301 | Shrimp (scampi) macrobactium frozen | Free | | |
| 03061302 | AFD shrimp frozen | Free | | |
| 03061303 | Prawns frozen | Free | | |
| 03061400 | Crabs | Free | | |
| 03061900 | Other, including flours, meals and pellets of crustaceans, fit for human | Free | | |

*Source:* Archives, Ministry of Commerce, Government of India

# Data Construction

1. QR removal: Identify affected products
   - More precise treatment assignment
   - $\sim$ 200 industries in past work vs. 10k+ products
2. QR removal: Identify affected firms

## Novel Product Concordance

| Firm Data | | Customs Notifications (HS-8) | |
|---|---|---|---|
| 3008040800 | Sunflower seed oil | 15121910 | Sunflower oil edible grade |
| 3008040804 | Sunflower seed oil, refined | | |
| 4012080400 | Suitcases | 42021204 | Plastic moulded suit-cases |
| 5024200404 | Distempers | 32100001 | Distempers |
| 6308361216 | Washing Machines/Laundry Mach | 84501100 | Fully - automatic washing machines (upto 10kg) |
| | | 84501200 | Other washing machines with built-in centrifugal drier (upto 10kg) |
| | | 84501300 | Other washing machines (up to 10kg) |

**6,000 products reported by Prowess firms**

$\longleftrightarrow$

**10,000 8-digit HS products in customs notifications**

# Data Construction

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2. QR removal: Identify affected firms
3. Impact (financial): CMIE Prowess:
   - Product scope
   - Balance sheet + financial statements Summary Stats

# Data Construction

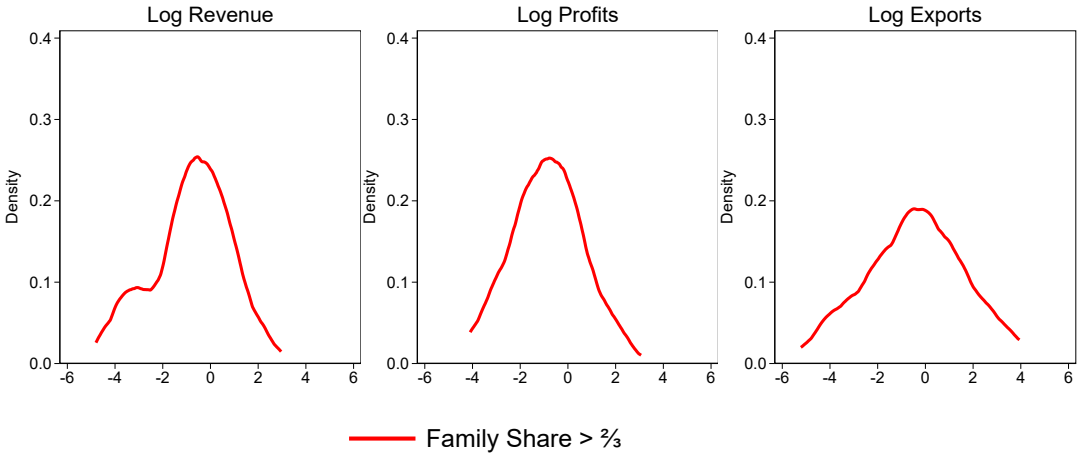
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4. Impact (managerial): MCA Admin Data
   - Family ties among top execs/directors
   - Tenure history from 1980s
   - $>$ 6 million directors

**Family Ties Among Top Executives**

| Name | Father's Name | Executive Director | Sex | Age | Place of Birth |
|---|---|---|---|---|---|
| A Khosla | D K Khosla | Yes | M | 44 | Delhi |
| M Khosla | D K Khosla | Yes | M | 40 | Delhi |
| P Khosla | D K Khosla | Yes | M | 39 | Delhi |
| N Khosla | N K Khosla | Yes | M | 38 | Delhi |
| D K Khosla | K L Khosla | Yes | M | 72 | Delhi |
| N K Khosla | K L Khosla | Yes | M | 69 | Jhelum |
| M P Gupta | P D Gupta | No | M | 70 | Delhi |
| V K Sood | H R Sood | No | M | 67 | Lahore |
| M L Mangla | T Chand | No | M | 70 | Sangrur |

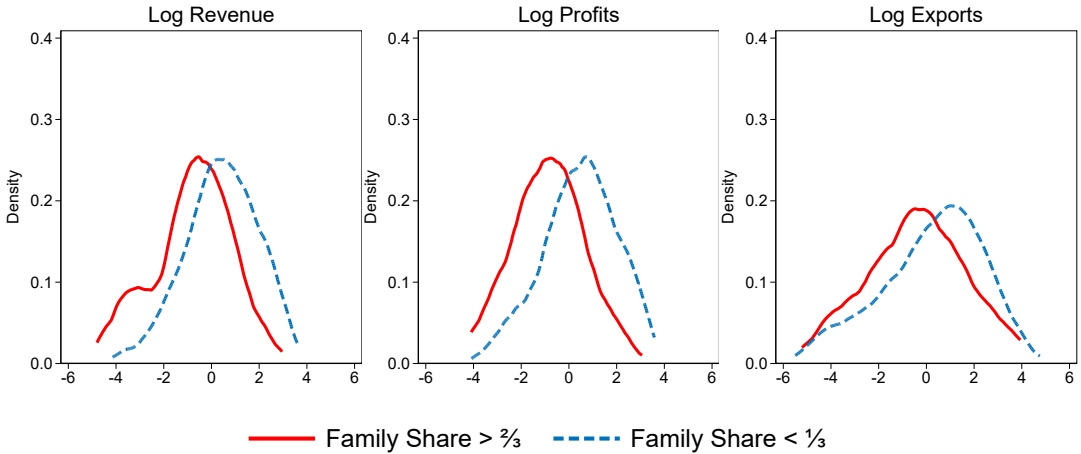
*Note:* Names abbreviated and anonymized. Based on a real firm.
Source: Confidential administrative records, Corporate Affairs Ministry

# Baseline Descriptive Statistics: Family Firms are Smaller and Less Profitable




Model: Baseline Equilibrium | Productivity Distribution | Estimation

# Baseline Descriptive Statistics: Family Firms are Smaller and Less Profitable



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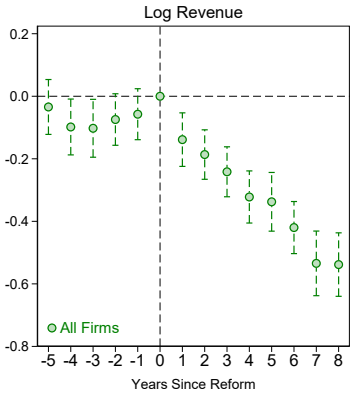
# Event-study Design

$$Y_{it} = \sum_{k=\underline{T}}^{\overline{T}} \theta_k D_{it}^k + \delta_i + \lambda_{jt} + \varepsilon_{it}$$

- $D_{it}^k := \mathbb{1}[t = s_i + k]$ are event-time dummies
  - $s_i$: first year in which QRs are removed on firm $i$'s highest-revenue (8-digit) product
  - Robust to alternate treatment assignment rules
- $\delta_i$: firm FE
- $\lambda_{jt}$: year $\times$ 3-digit industry FE

- Sun and Abraham (2021) estimator. Only using never-treated firms as controls

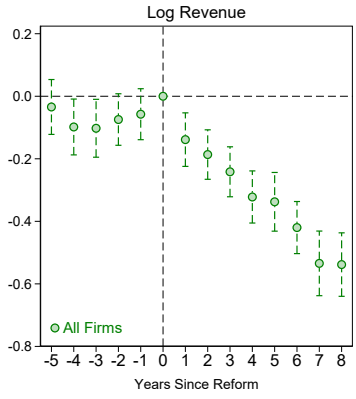
# ❶ First Stage: Firms Contract

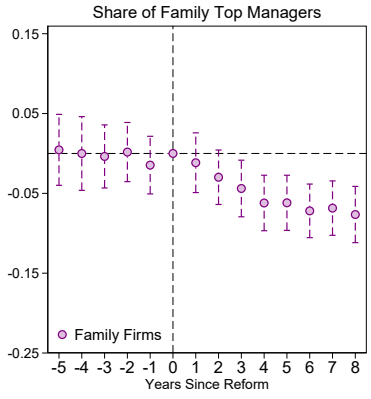



Treated vs. Control Group | Wage Bill and Inputs | Alt Specs | Fam/Prof Firms Separately | Bounds Financial | Bounds Management | Survivors

Number of Managers | Family Majority | Place and State of Birth | Directors' Age Distribution | Ownership | Exit | Model: Baseline Equilibrium | Model: Comparative Statics

# ❶ First Stage: Firms Contract ❷ Laggards Professionalize

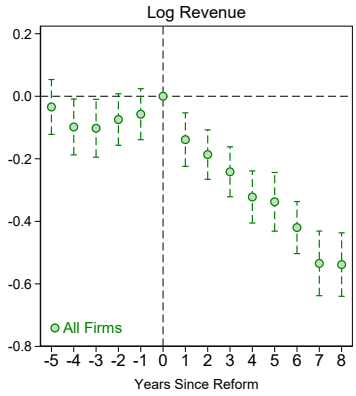


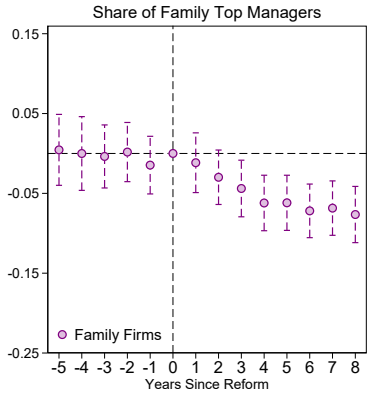



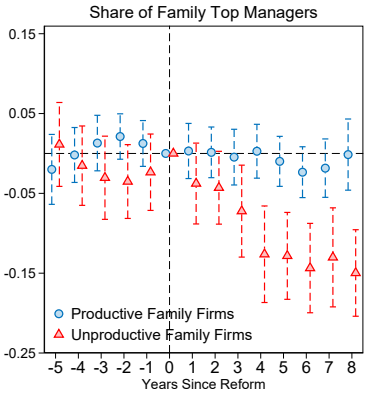

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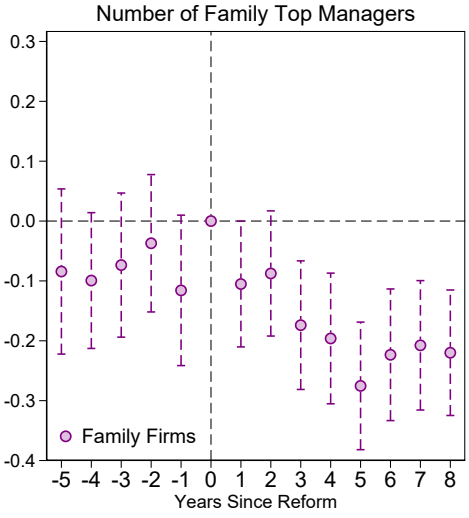



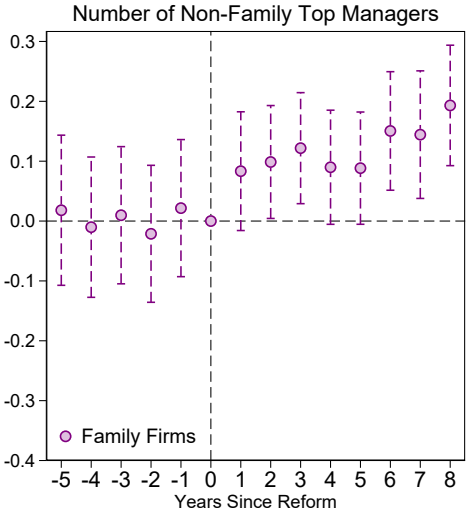




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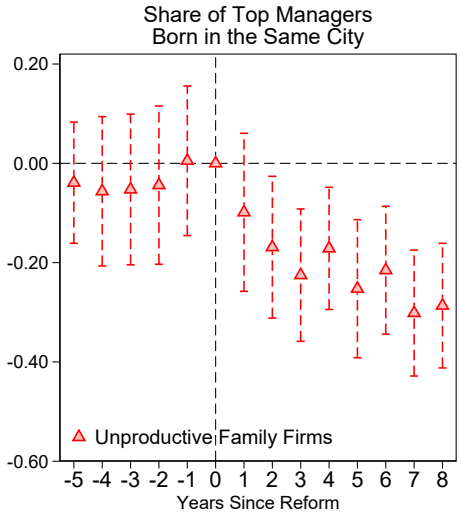
# ❷ Laggards Professionalize: Number of Family and Professional Executives

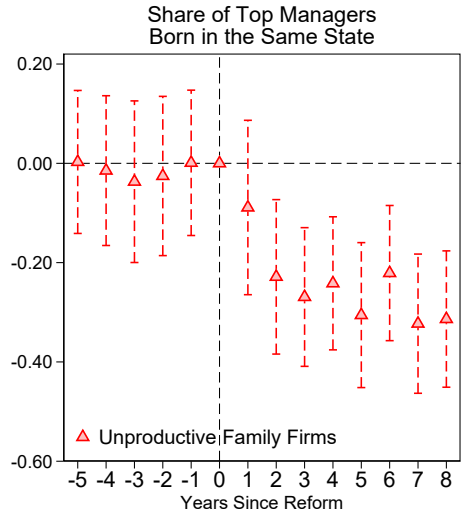



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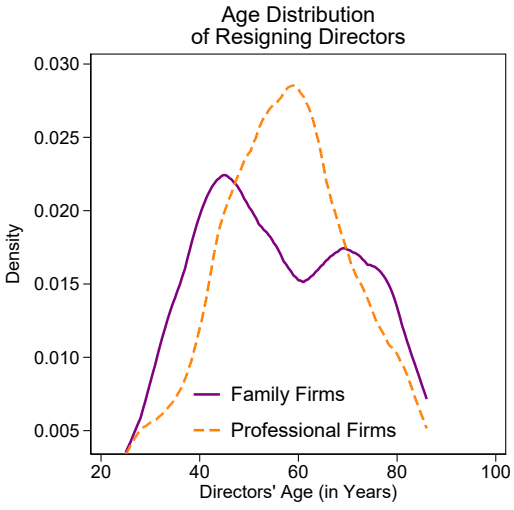
# ❷ Laggards Professionalize: City and State of Birth

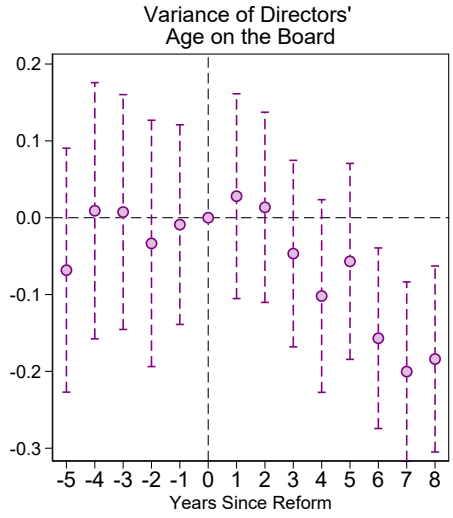



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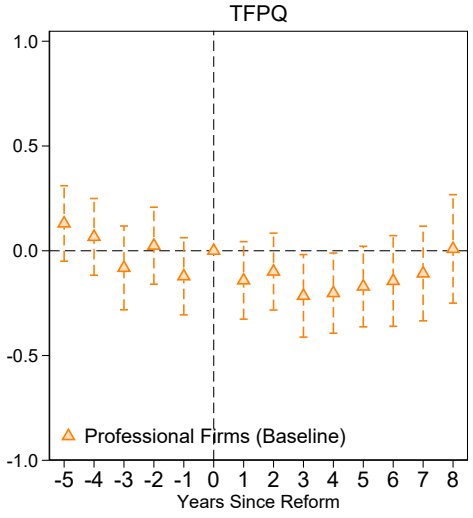
# ❷ Laggards Professionalize: Compression of the Executives' Age Distribution



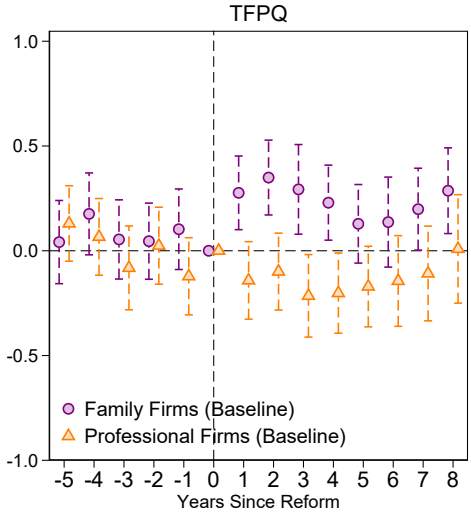




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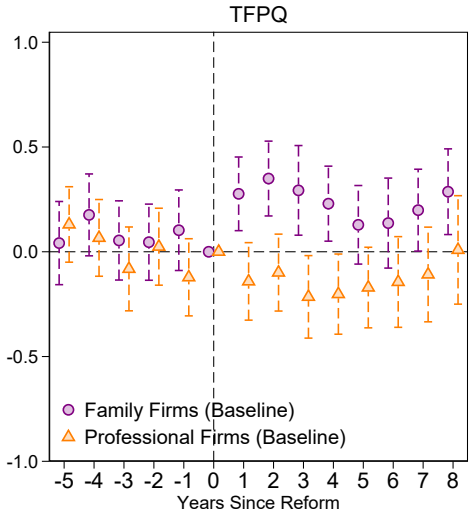
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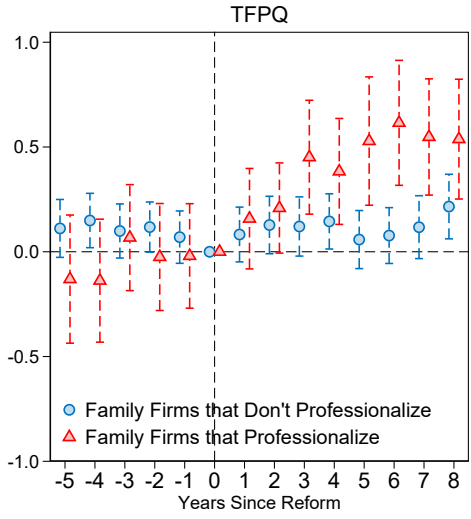


# ❸ Firms that Professionalize Report Higher Quantity Productivity



# ❸ Firms that Professionalize Report Higher Quantity Productivity




*Note:* A family firm is defined as having professionalized if its share of family top managers was lower in $t = 8$ as compared to $t = 0$.

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# Environment

$$U = X^{1-\alpha} Y^{\alpha}$$

- **Differentiated goods, $Y$:** CES demand and heterogeneous firms. Domestic firms sell at home and compete with imports
- **Homogeneous outside good, $X$:** Freely traded, competitively produced with a unit labor requirement. Pins $w = 1$; exports pay for differentiated imports

# Management Choice

- Monopolistic competition with CES demand, heterogeneous firm productivity, $z$

Preferences | Technology

$$\text{Entrants draw } z \sim G(z) \longrightarrow \text{Firm's payoff} = \begin{cases} \pi(z) + \mathcal{B} & \text{if firm stays family} \\ \pi(\gamma z) & \text{if firm professionalizes} \\ 0 & \text{if firm exits} \end{cases}$$

- $\mathcal{B}$: owners derive non-monetary private benefits from running a firm as a family firm
  (Demsetz and Lehn, 1985; Burkart, Panunzi, and Shleifer, 2003; Bertrand and Schoar, 2006; Hurst and Pugsley, 2015)
- $\gamma > 1$: gains from professionalizing firm management. Absorbing state — Reputation Cost
- Firm owners are hand-to-mouth: $\implies$ exit if $\pi(z) < 0$

Equilibrium

# Management Choice

- Monopolistic competition with CES demand, heterogeneous firm productivity, $z$

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$$\text{Entrants draw } z \sim G(z) \longrightarrow \text{Firm's payoff} = \begin{cases} \pi(z) + \mathcal{B} & \text{if firm stays family} \\ (1-\tau) \cdot \pi(\gamma z) - f_d & \text{if firm professionalizes} \\ 0 & \text{if firm exits} \end{cases}$$

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- Firm owners are hand-to-mouth: $\implies$ exit if $\pi(z) < 0$

Equilibrium

# Management Choice

- Monopolistic competition with CES demand, heterogeneous firm productivity, $z$

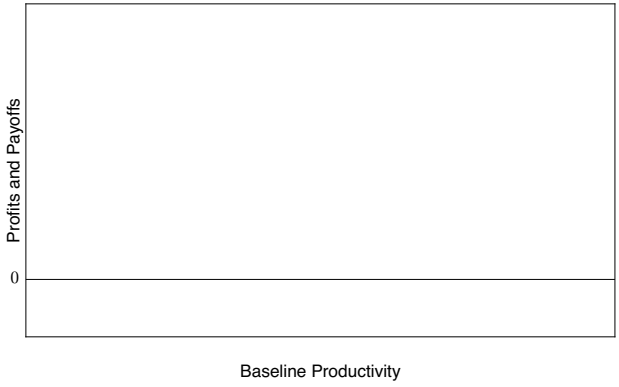
Preferences | Technology

$$\text{Entrants draw } z \sim G(z) \longrightarrow \text{Firm's payoff} = \begin{cases} \pi(z) + \mathcal{B} & \text{if firm stays family} \\ \pi(\gamma z) & \text{if firm professionalizes} \\ 0 & \text{if firm exits} \end{cases}$$

- $\mathcal{B}$: owners derive non-monetary private benefits from running a firm as a family firm
  (Demsetz and Lehn, 1985; Burkart, Panunzi, and Shleifer, 2003; Bertrand and Schoar, 2006; Hurst and Pugsley, 2015)
- $\gamma > 1$: gains from professionalizing firm management. Absorbing state — Reputation Cost
- Firm owners are hand-to-mouth: $\implies$ exit if $\pi(z) < 0$

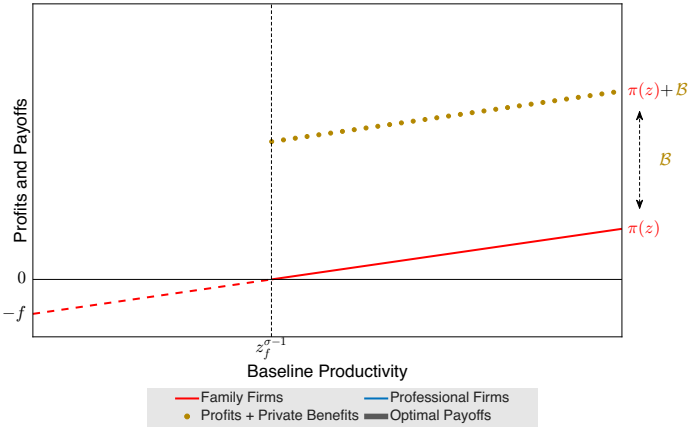
Equilibrium

# Firm Profits and Management Choice:




Note: Baseline productivity: before decision to professionalize.

# Firm Profits and Management Choice: Family Firms



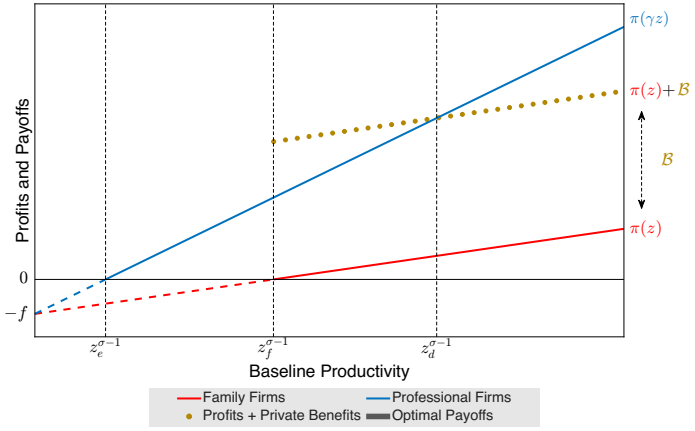

## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

# Firm Profits and Management Choice: Professional Firms




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

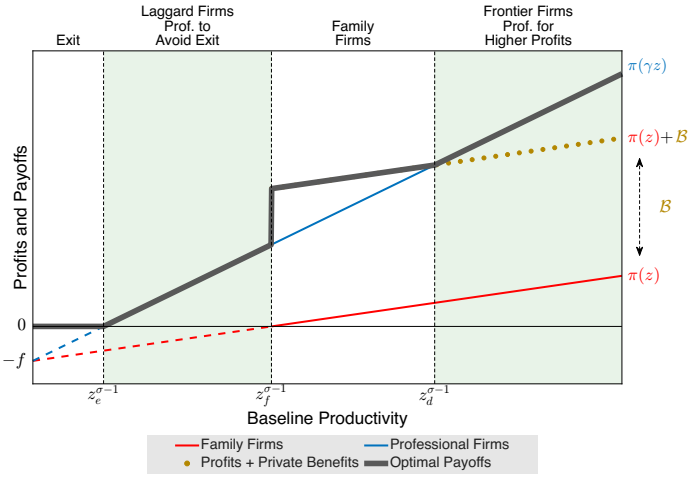
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

# Firm Profits and Management Choice: Optimal Payoff




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

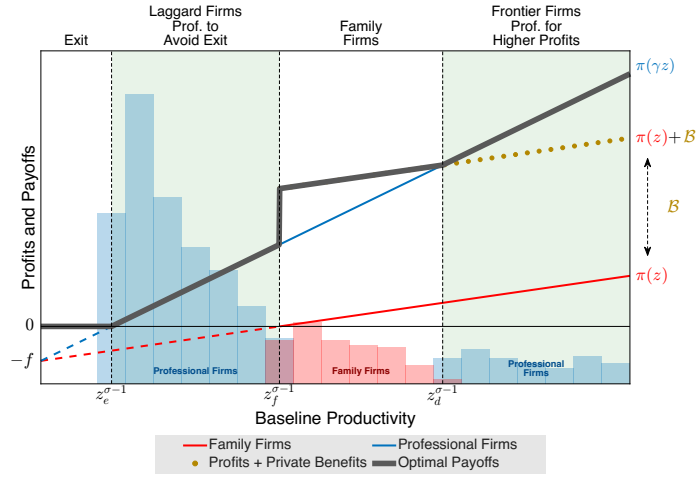
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before decision to professionalize.

Exit | Contracting Frictions | Productivity Distribution | Data | Event Studies | Estimation

# Firm Profits and Management Choice: Baseline Productivity Distribution




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

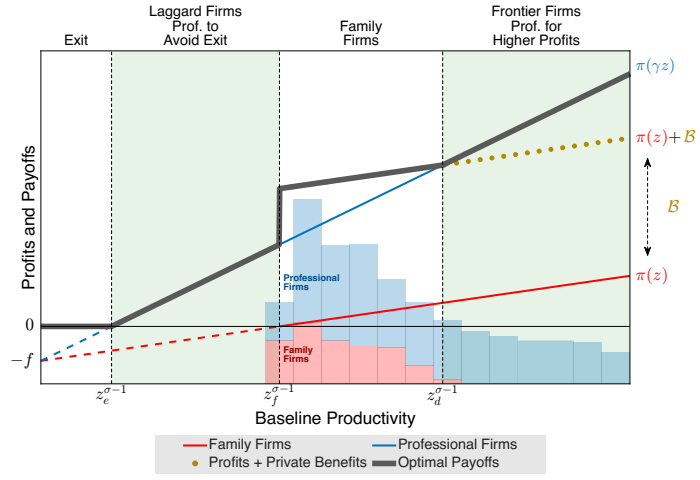
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Model: Baseline Equilibrium | Data

# Firm Profits and Management Choice: Observed Productivity Distribution




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

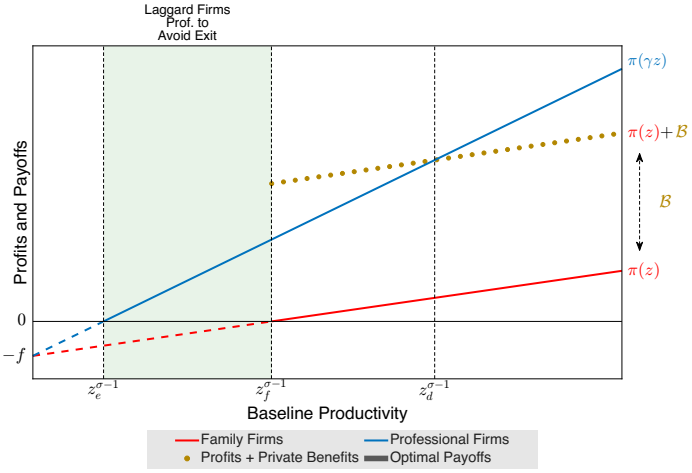
Note: Baseline productivity: before the decision to professionalize.

Model: Baseline Equilibrium | Data

# Import Competition

- **Shock:** exogenous increase foreign varieties available in India, $M_F \uparrow$
- **Domestic adjustment:** $M_F \uparrow \Longrightarrow P \downarrow \Longrightarrow A \propto EP^{\sigma-1} \downarrow$
  - firm profits $\downarrow$
  - firms exit or professionalize management
- Short run, fixed entry equilibrium

# Family Firms: Response to ↑ Import Competition, ↓ Market Demand




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

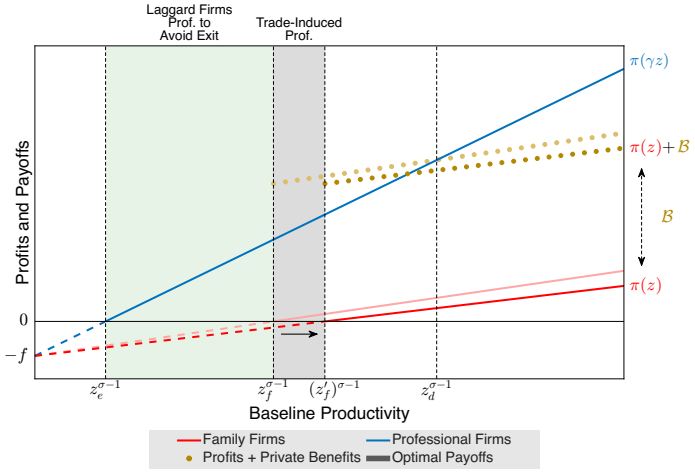
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f\left(\gamma^{\sigma-1}-1\right)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

# Family Firms: Response to ↑ Import Competition, ↓ Market Demand




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

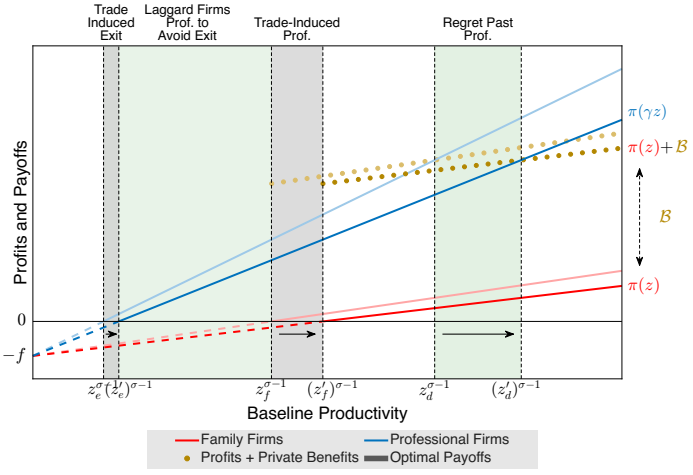
$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

# Professional Firms: Response to ↑ Import Competition, ↓ Market Demand




## Productivity Thresholds

$$z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$$

$$z_e = \frac{1}{\gamma} z_f$$

$$z_d = \left[\frac{\mathcal{B}}{f(\gamma^{\sigma-1}-1)}\right]^{\frac{1}{\sigma-1}} \cdot z_f$$

Note: Baseline productivity: before the decision to professionalize.

Import Competition | Contracting Frictions | Reputation Cost | Event Studies

# Outline

1. Introduction
2. Natural Experiment
3. Data
4. Empirical Results
5. Model
6. **Calibration & Counterfactuals**

# Calibration

$$\Theta = \{\sigma, k, \gamma, \mathcal{B}\}$$

- $\sigma = 4$
- $k = 3.5 \longrightarrow$ upper-tail slope in a log-rank vs. log-productivity QQ plot Head, Mayer, Thoenig (2014)
- $\mathcal{B}, \gamma$ jointly determine management choice Moment Conditions
- Calibrate the model to India's pre-policy economy

| Parameter | Identifying moment | Estimate |
|---|---|---|
| $\mathcal{B}$ | Share of family firms | 0.18 |
| $\gamma$ | Log mean revenue gap: professional minus family firms | 1.23 |

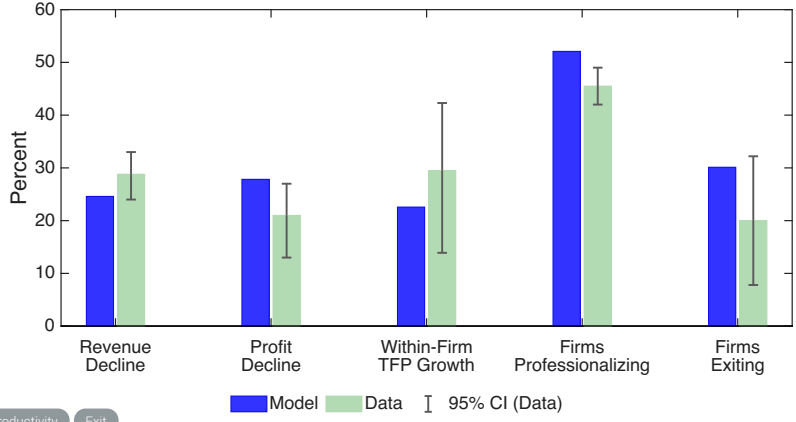
Data Model: Baseline Equilibrium

# Comparison of Model and Data Estimates

- **Policy experiment:** increase import penetration from its pre- to post-policy level
- Re-solve the post-shock allocation
- Compare model-implied changes in untargeted moments to event-study results

# Comparison of Model and Data Estimates

- **Policy experiment:** increase import penetration from its pre- to post-policy level
- Re-solve the post-shock allocation
- Compare model-implied changes in untargeted moments to event-study results




Revenue and Profits | Productivity | Exit

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) – (2) |
|---|---|---|---|
| ***Panel A. Productivity change*** | | | |
| Total | | | |
| Between-firm selection | | | |
| Within-firm productivity | | | |
| Domestic firm mass | | | |
| Imports | | | |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | | | |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) – (2) |
|---|---|---|---|
| ***Panel A. Productivity change*** | | | |
| Total | 8.67 | | |
| Between-firm selection | 8.07 | | |
| Within-firm productivity | 0.60 | | |
| Domestic firm mass | | | |
| Imports | | | |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | | | |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) − (2) |
|---|---|---|---|
| ***Panel A. Productivity change*** | | | |
| Total | 8.67 | 11.65 | −2.98 |
| Between-firm selection | 8.07 | 11.65 | −3.58 |
| Within-firm productivity | 0.60 | 0.00 | +0.60 |
| Domestic firm mass | | | |
| Imports | | | |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | | | |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

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| ***Panel A. Productivity change*** | | | |
| Total | 8.67 | 11.65 | −2.98 |
| Between-firm selection | 8.07 | 11.65 | −3.58 |
| Within-firm productivity | 0.60 | 0.00 | +0.60 |
| Domestic firm mass | −9.50 | −13.53 | +4.02 |
| Imports | | | |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | | | |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) − (2) |
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| Domestic firm mass | −9.50 | −13.53 | +4.02 |
| Imports | 8.97 | 8.97 | 0.00 |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | | | |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) − (2) |
|---|---|---|---|
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| Within-firm productivity | 0.60 | 0.00 | +0.60 |
| Domestic firm mass | −9.50 | −13.53 | +4.02 |
| Imports | 8.97 | 8.97 | 0.00 |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | 8.14 | 7.10 | +1.05 |
| Welfare including private benefits | | | |

Details

# Productivity and Welfare

| | Benchmark model Professionalization permitted | Counterfactual No new professionalization | Difference (1) − (2) |
|---|---|---|---|
| ***Panel A. Productivity change*** | | | |
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| Between-firm selection | 8.07 | 11.65 | −3.58 |
| Within-firm productivity | 0.60 | 0.00 | +0.60 |
| Domestic firm mass | −9.50 | −13.53 | +4.02 |
| Imports | 8.97 | 8.97 | 0.00 |
| ***Panel B. Welfare change relative to baseline*** | | | |
| Consumption welfare | 8.14 | 7.10 | +1.05 |
| Welfare including private benefits | 7.47 | 6.50 | +0.97 |

Details

# Conclusion

- Trade drives aggregate growth by improving laggard firms' productivity
- Empirical lens here is family firms, the mechanism is more general
- Protection from competition can sustain taste-based deviations from merit (Becker 1957)
  - along multiple dimensions: kinship, caste, religion, race, or gender
- Competition may help restore meritocracy
- These organizational changes occur without deeper institutional shifts

Thank you!

**Ananya Kotia**

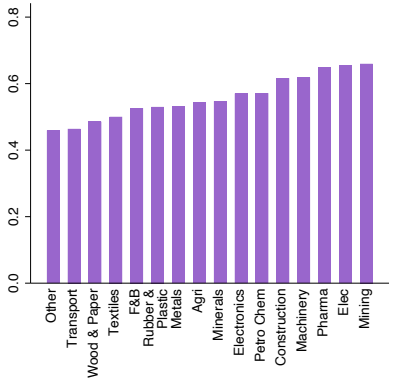
Stanford University

*ananyakotia.com*

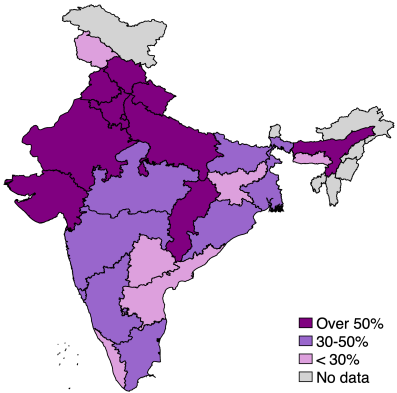
# Family Firms are Widespread in India

Family Firms: at least two family members on board

Share of Family Firms across Industries




Share of Family Firms across States

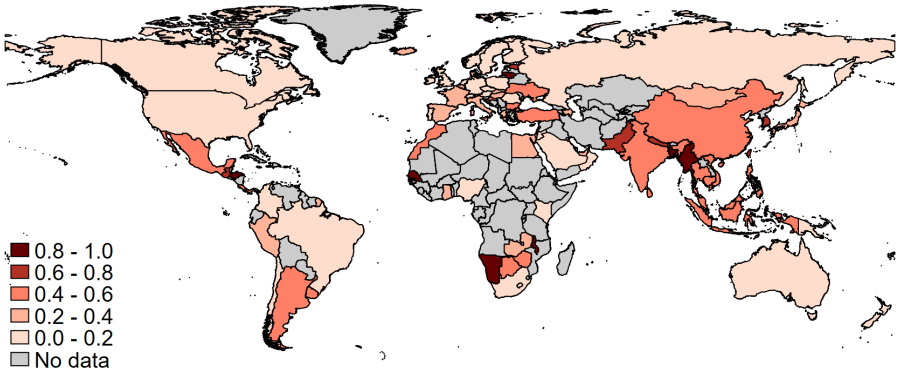



*Source:* Ministry of Corporate Affairs and CMIE Prowess$_{dx}$

Back

# Family Firms are Widespread in Developing Countries

## Share of Directors Sharing a Surname




*Source:* BoardEx, Wharton Research Data Service (WRDS)

Back

# Indian Response to WTO's QR Ruling

*"The commerce secretary had written to me to say that they intended to make a case for continuing with QRs on consumer goods for BoP reasons, and he wanted confirmation of support from the Ministry of Finance. I promptly wrote back to say that* ***it would be against our national interest because it painted a picture of helplessness that was completely contrary to what we in the Ministry of Finance were projecting to foreign investors****"*

—Montek Singh Ahluwalia, Former Indian Finance Secretary and Deputy Chairman of the Planning Commission

Policy response by India:

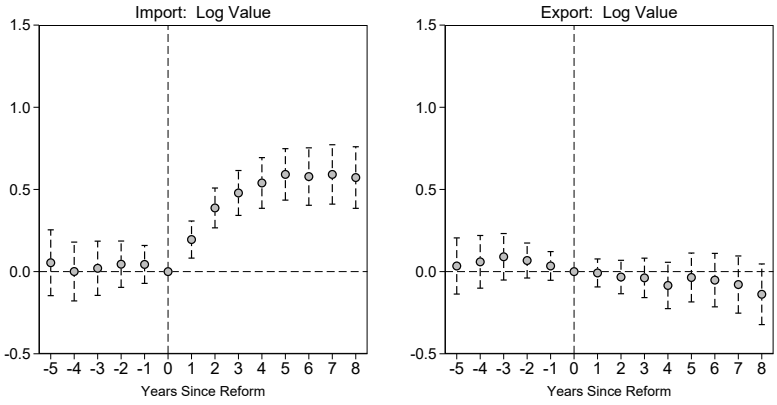
- Tariffs maintained : limited room to raise beyond WTO bindings
- Canalization of Grains
- "War Room" & sensitive list: DGFT monitored ~300 consumer goods
- Some anti-dumping (esp. chemicals, steel, consumer goods)

Back

# HS Codes Example: Electrical machinery

- **8517:** Telephone sets and other apparatus for voice transmission
  - **861711:** Line telephone sets
    - **85171110:** Push button type
    - **85171120:** Rotary dial type
    - **85171130:** Cordless Telephones
  - **861712:** Cellular network sets
    - **85171210:** Push button type
- **8525:** Transmission apparatus and cameras
  - **852550:** Transmission apparatus
    - **85255010:** Radio broadcast transmitter
    - **85255020:** TV broadcast transmitter
    - **85255040:** Jamming equipment
  - **852580:** Cameras
    - **85258010:** Television Cameras
    - **85258020:** Digital cameras
    - **85258030:** Video cameras recorders

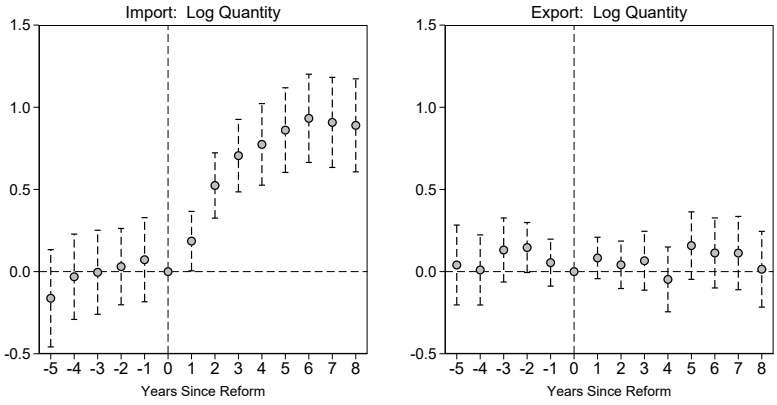
# QR-Removal: Imports Soar, No Effect on Exports




$$y_{pt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{pt}^k + \delta_p + \lambda_{qt} + \varepsilon_{pt}$$

$p$: 6-digit HS product, $q$: 4-digit HS product, $D_{pt}^k$: event-time dummies Quantities Back

# QR-Removal: Imports Soar, No Effect on Exports




$$y_{pt} = \sum_{k=\underline{T}}^{\bar{T}} \beta_k D_{pt}^k + \delta_p + \lambda_{qt} + \varepsilon_{pt}$$

$p$: 6-digit HS product, $q$: 4-digit HS product, $D_{pt}^k$: event-time dummies Timeline Values

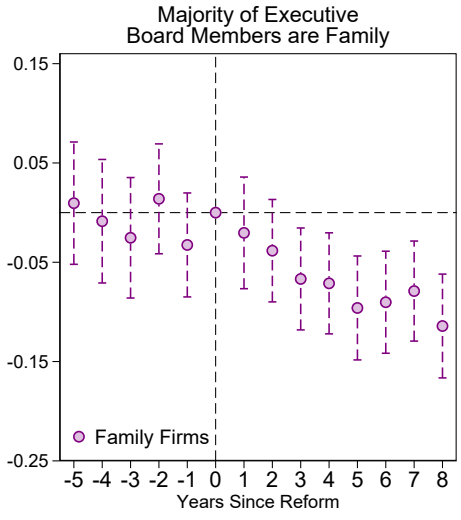
# Summary Statistics

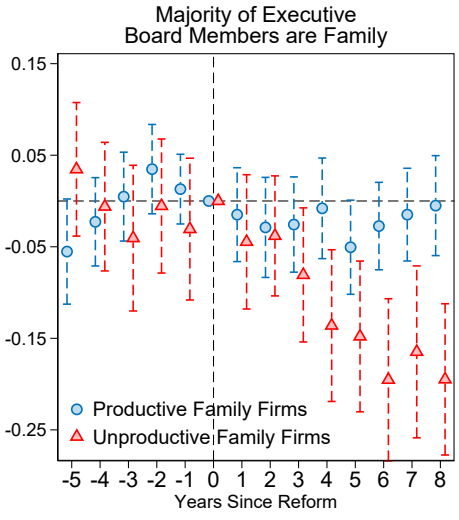
| | Obs | p10 | Mean | p50 | SD |
|---|---|---|---|---|---|
| Treated Firms (%) | 4,996 | 0 | 0 | 1 | 0 |
| Firm Age (Years) | 83,726 | 7 | 26 | 56 | 20 |
| Wages | 77,724 | 1 | 220 | 326 | 1,382 |
| Total Assets | 82,465 | 46 | 4,083 | 4,983 | 31,652 |
| Revenues | 76,552 | 19 | 3,674 | 4,785 | 36,687 |
| Expenses on Raw Materials | 67,944 | 13 | 1,687 | 2,227 | 17,783 |
| At least 2 Family Members on Board | 4,852 | 0.00 | 0.45 | 1.00 | 0.50 |
| Family Share on Board | 39,644 | 0.10 | 0.38 | 0.75 | 0.25 |
| Family Share on Executive Board | 39,644 | 0.00 | 0.64 | 1.00 | 0.42 |

*Note:* All nominal variables are reported in INR million and are deflated to 2005 prices using the GDP deflator.

Back

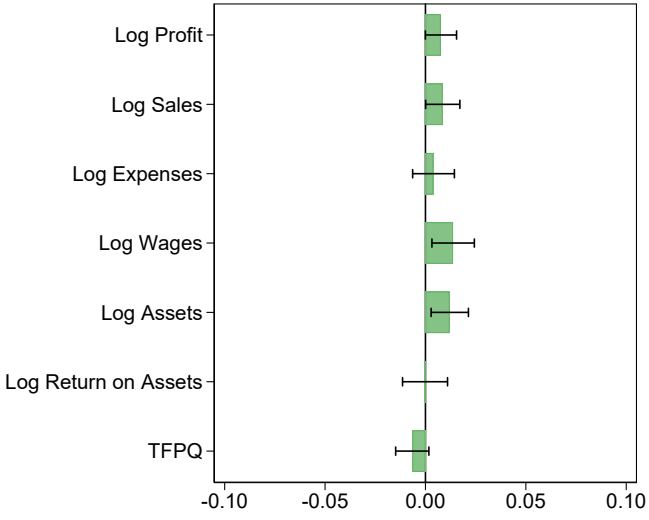
# Family Majority on the Executive Board



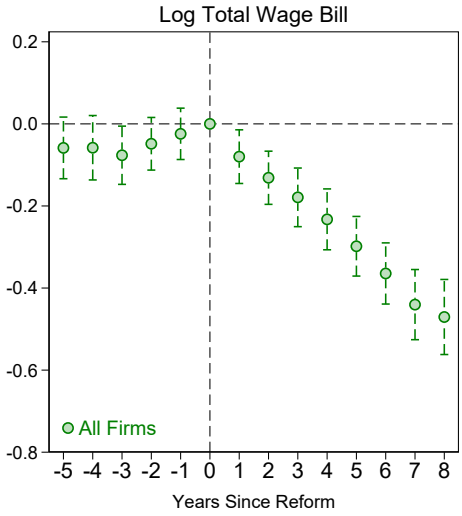

Back

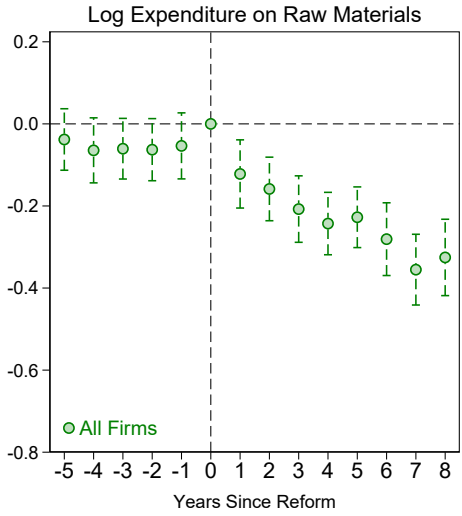
# Treated and Control Firms are Similar




Back

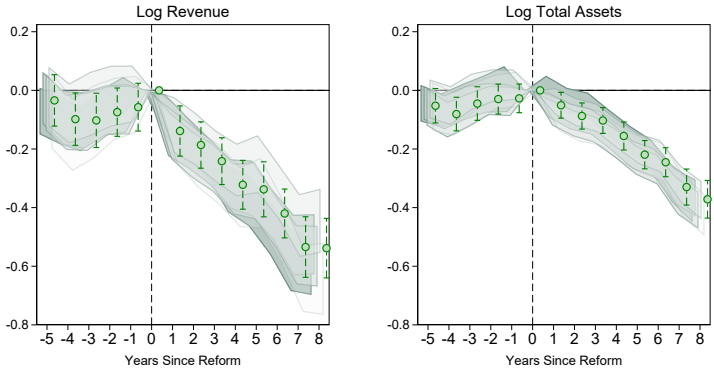
# ❶ All Firms Contract: Total Wage Bill and Expenses




Back

# ❶ All Firms Contract: Robustness to Alternative Specifications




Alternative Fixed Effects:

(1) Firm + Year

(2) Firm + (Industry × Year)

(3) Firm + (Industry × Year) + (State × Year)

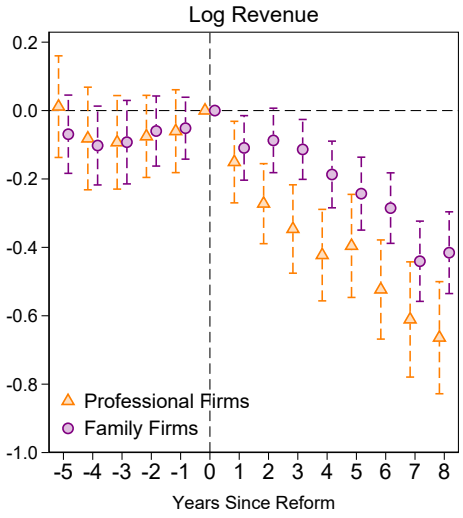
(4) Firm + (Industry × Year) + (District × Year)

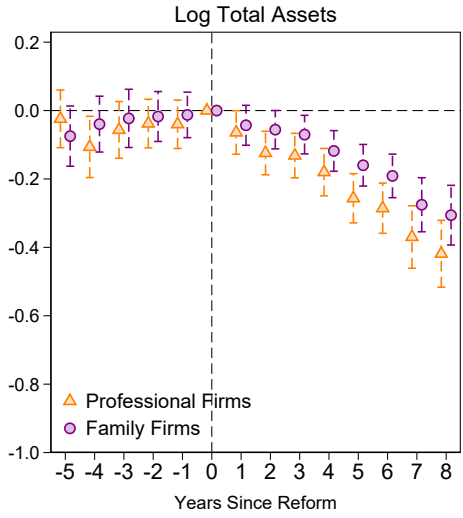
(5) Firm + (Industry × Year × State)

(6) Firm + (Industry × Year × District)

Back

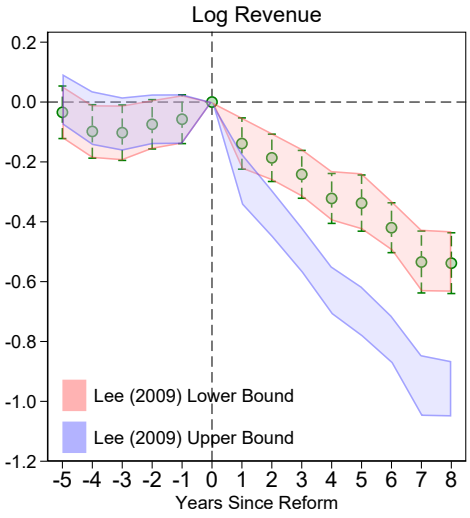
# ❶ All Firms Contract

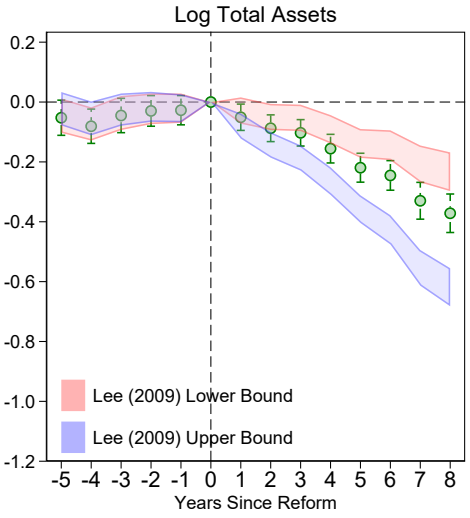



Back

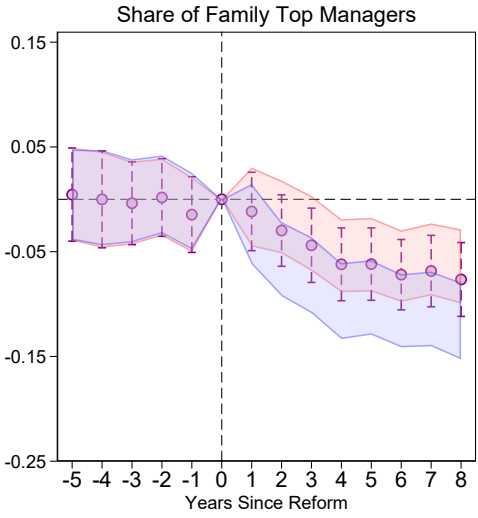
# Lee Bounds



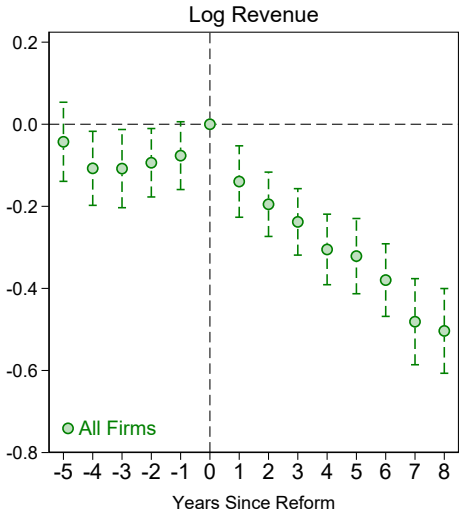

Back

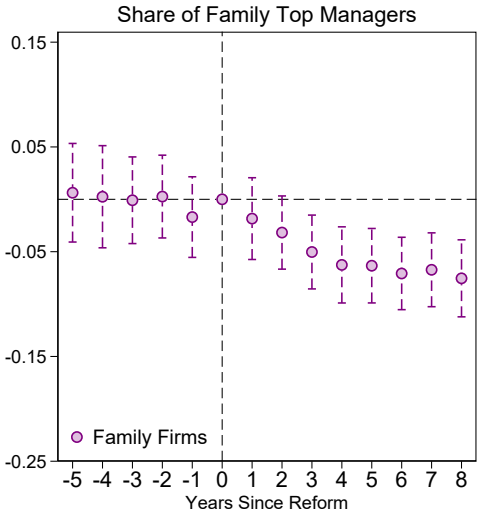
# Lee Bounds




Back

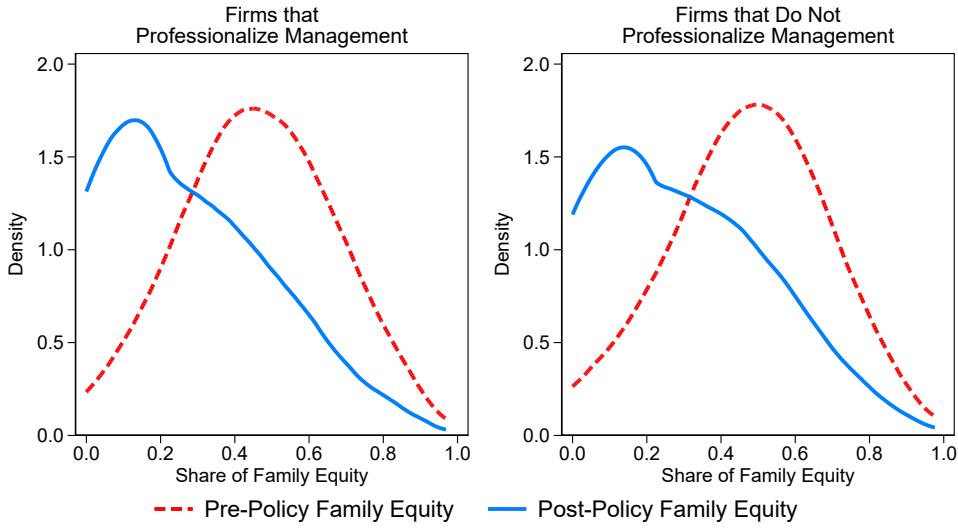
# Conditioning on Survivors




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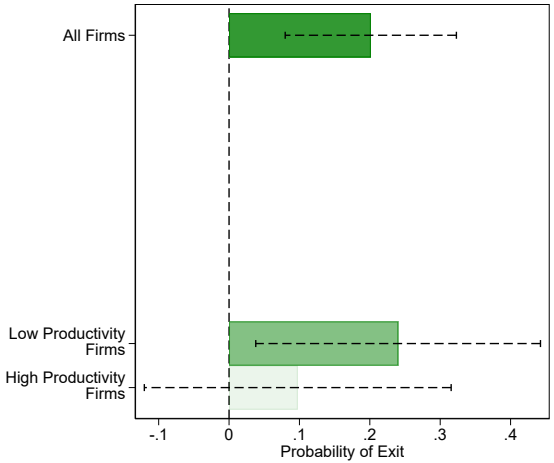
# Family Ownership




Back

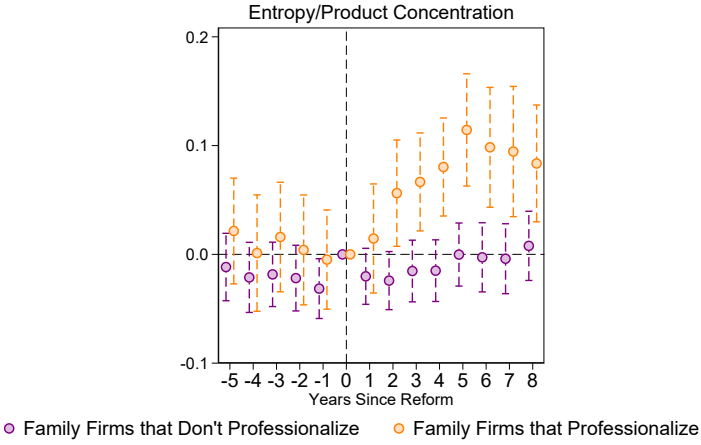
# Exit (in the Long Run)

- Cannot confirm exit status or exit year in Prowess data
- Verify exit status by matching firms that drop out of Prowess to MCA admin data
- Long-difference: measure whether firm exits in any year from 2001–2021




Back Model vs Data

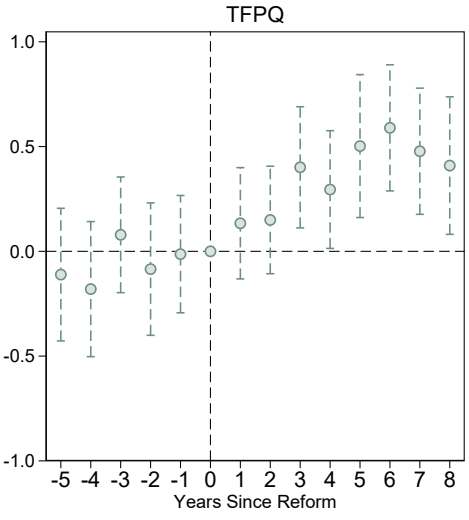
# ❸ Firms that Professionalize Report Higher Product Concentration




*Note:* A family firm is defined as having professionalized if its share of family top managers was lower in $t = 8$ as compared to $t = 0$.

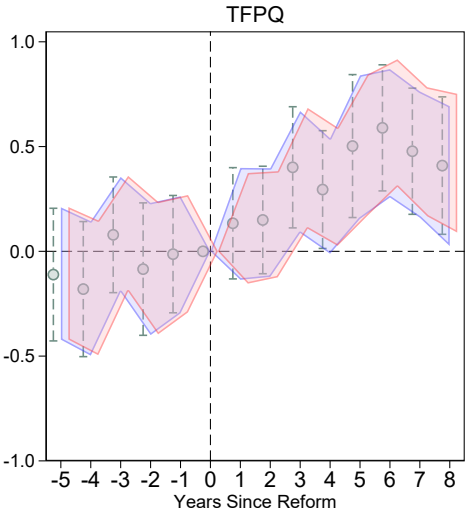
Back

# Firms that Professionalize Report Higher Productivity (Triple Difference)



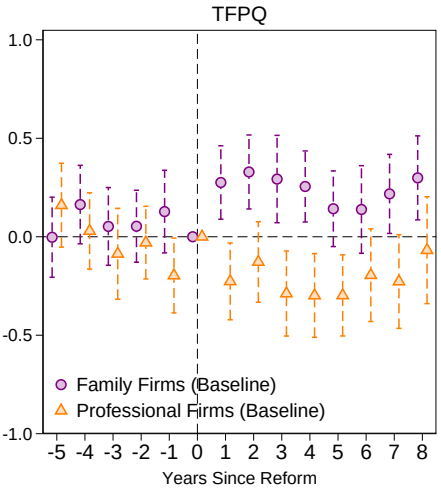

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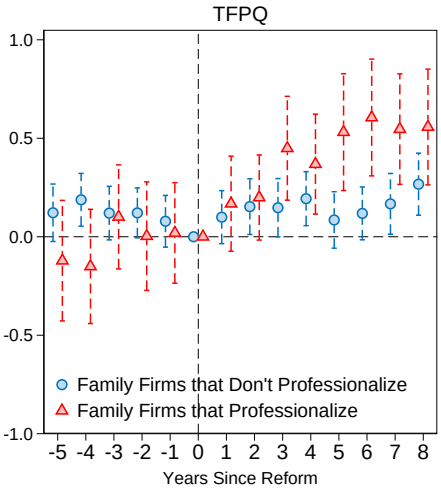
# Lee Bounds




Back

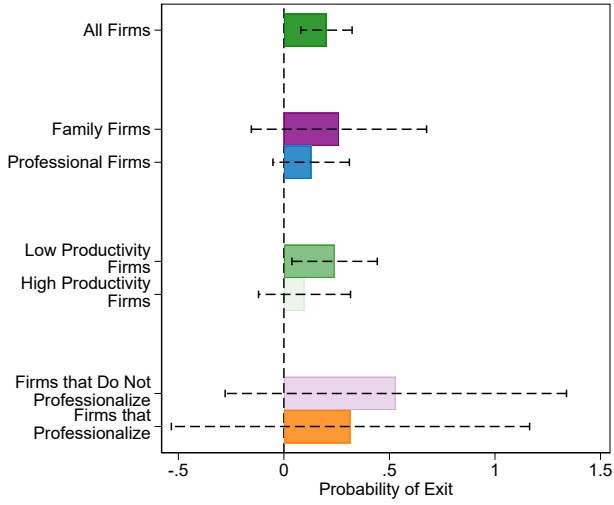
# Quantity Productivity: Conditioning on Survivors



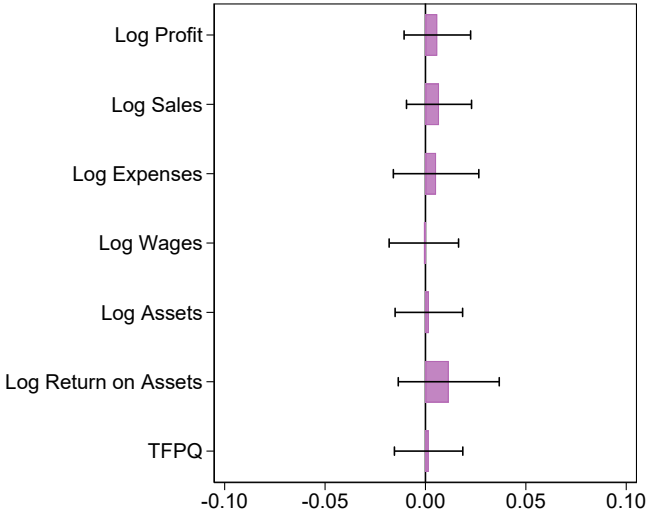




Back

# Exit


All Firms
Family Firms
Professional Firms
Low Productivity Firms
High Productivity Firms
Firms that Do Not Professionalize
Firms that Professionalize
-.5
0
.5
1
1.5
Probability of Exit

# Selection Into Professionalizing Management




Back

# Preferences

- Representative consumer

$$\mathcal{U} = X^{1-\alpha} Y^{\alpha} \qquad (0 < \alpha < 1)$$

  - $X$: freely-traded homogeneous good, with unit labor requirement
  - $Y$ is a CES aggregate of differentiated varieties with elasticity of substitution $\sigma > 1$
- The mass of foreign varieties is $M_F$
- Consumers spend share $\alpha$ of total expenditure $E$ on good $Y$
- Demand for each variety $y$ is given by $y = \alpha E P^{\sigma-1} p^{-\sigma}$

# Technology

- Firms differ in productivity $z$ and produce differentiated varieties using labor $\ell$.
- Constant marginal cost with a fixed overhead cost: $\ell(z) = f + \frac{y}{z}$
- Prices are a markup over marginal cost: $p(z) = \frac{\sigma}{\sigma-1} \cdot \frac{w}{z}$
- Profits: $\pi(z) = Az^{\sigma-1} - wf$ $\left(\text{where } A = \frac{1}{\sigma}\left(\frac{\sigma-1}{\sigma}\right)^{\sigma-1} EP^{\sigma-1}w^{1-\sigma} = \text{market demand}\right)$

Management Choice

# Reputational Cost of Switching Back to Family Management



Reuters

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## Infosys CEO resigns after long-running feud with founders

By **Samantha Kareen Nair** and **Sankalp Phartiyal**

August 18, 2017 3:58 PM GMT+1 · Updated 8 years ago

Aa

BENGALURU/MUMBAI (Reuters) - Vishal Sikka, the chief executive brought in to turn around India's Infosys three years ago, resigned on Friday, blaming a "continuous drumbeat of distractions" and a long-running row with the founders over company strategy.

Sikka's resignation spooked investors in India's second-biggest IT services company and sent its shares down nearly 10 percent, wiping $3.45 billion off its market value. The stock touched its lowest level since the start of Sikka's tenure.

Model: Management Choice | Model: Comparative Statics | Profits After Trade

# Profits After Trade Liberalization ($t = 1$)

- Switching from professional to family management incurs (reputational) cost $\kappa$ Example

$$\text{Payoff}\,(z, \mathcal{P}_0, \mathcal{P}_1) = \mathcal{P}_1 \pi(\gamma z) + (1 - \mathcal{P}_1)\left[\pi(z) + B - \kappa \mathcal{P}_0\right]$$

- When $\mathcal{P}_1 = 1$ (the firm upgrades management in $t = 1$), the payoff is $\pi(\gamma z)$ regardless of the past choice
- When $\mathcal{P}_1 = 0$ (the firm is family-managed in $t = 1$), the payoff is:
  - $\pi(z) + B$ if it was family in $t = 0$ (i.e., $\mathcal{P}_0 = 0$)
  - $\pi(z) + B - \kappa$ if it was professionally-managed in $t = 0$ (i.e., $\mathcal{P}_0 = 1$)

Model: Baseline Equilibrium | Model: Comparative Statics | Model: Management Choice

# Equilibrium: Free Entry, Exit, and Equilibrium Price Index

- **Free entry:**

$$[1 - G(z_e)] \cdot \frac{1}{\delta}\pi - f_E = 0$$

- **Average profits among surviving firms:**

$$\pi = \frac{\sigma - 1}{k - \sigma + 1} f \cdot \Delta, \qquad \Delta \equiv 1 + \frac{k}{\sigma - 1}\left(\gamma^{\sigma-1} - 1\right)\gamma^{-k}\left(\left(\frac{\mathcal{B}}{f\left(\gamma^{\sigma-1} - 1\right)}\right)^{\frac{\sigma-k-1}{\sigma-1}} - 1\right)$$

- **Exit cutoff:**

$$z_e = \left(\frac{1}{\delta f_E} \cdot \frac{\sigma - 1}{k - \sigma + 1} f\Delta\right)^{1/k}$$

- **Price index (domestic varieties):**

$$P = \frac{1}{\rho\gamma}\left(\sigma f_E\right)^{\frac{1}{\sigma-1}}\left(\frac{\sigma - 1}{k - \sigma + 1} \cdot \frac{f}{\delta f_E} \cdot \Delta\right)^{-1/k}, \qquad \rho \equiv \frac{\sigma - 1}{\sigma}$$

Back

# Equilibrium: Market Clearing and Aggregates

- **Masses of firms.** Let $M_E$ be the mass of entrants and $M$ the mass of active firms:

$$M = \frac{1 - G(z_e)}{\delta} M_E$$

- **Labor demand in differentiated sector.** With revenue $R$ and aggregate operating profits $\Pi$:

$$L_Y = \frac{R - \Pi}{w} + M_E f_E$$

  Free entry implies $\Pi = w M_E f_E$

- **Goods-market clearing.** Revenue equals expenditure on differentiated goods:

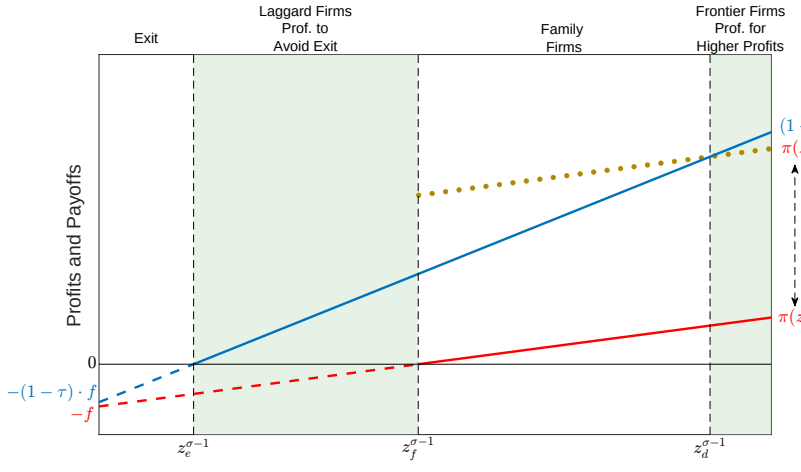
$$R = E, \qquad E = \alpha I$$

- **Numeraire and income.** Homogeneous good pins down $w = 1$ and total income is the wage bill:

$$I = wL = L \quad \Rightarrow \quad E = \alpha L, \;\; L_Y = \alpha L, \;\; L_X = (1 - \alpha) L$$

Back

# Contracting Frictions



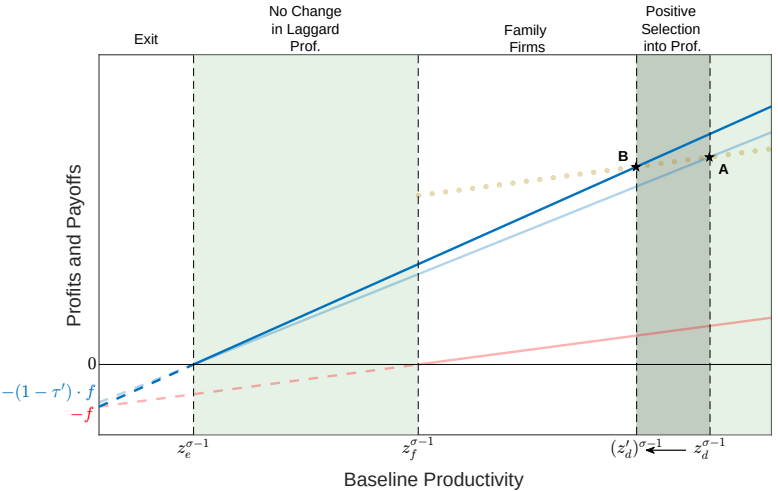

Productivity Thresholds:

- $z_f = \left(\frac{f}{A}\right)^{\frac{1}{\sigma-1}}$
- $z_e = \frac{1}{\gamma}\cdot z_f$
- $z_d = \left(\frac{\mathcal{B}-\tau f}{f\left((1-\tau)\gamma^{\sigma-1}-1\right)}\right)^{\frac{1}{\sigma-1}}\cdot z_f$

Note: Baseline productivity: before decision to professionalize.

Model: Baseline Equilibrium | Model: Comparative Statics

# Contracting Frictions



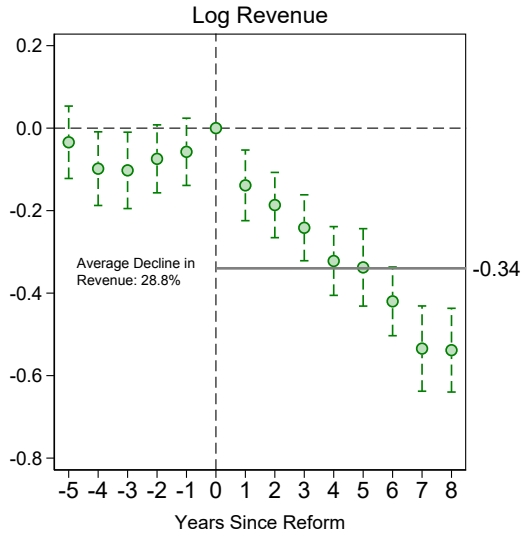

Productivity Thresholds:

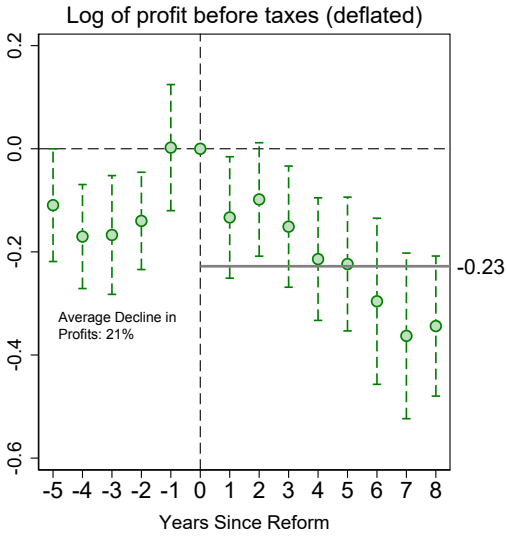
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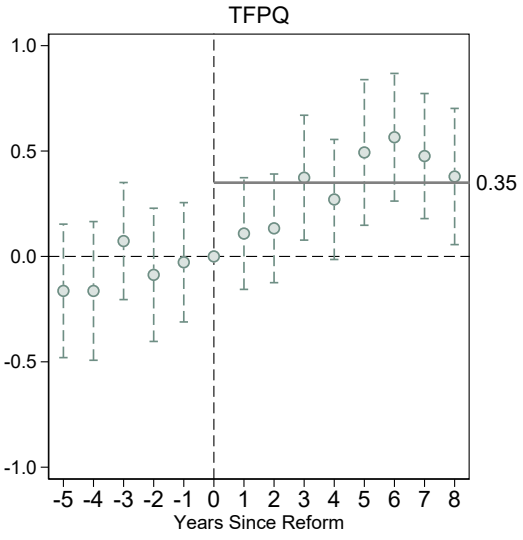
# Revenue and Profit Decline




Back

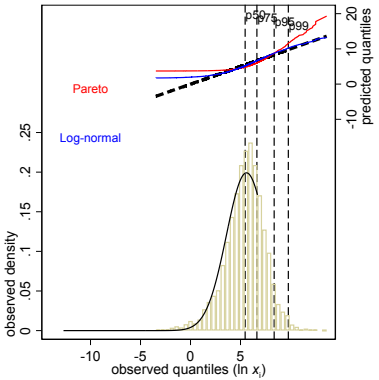
# Average Post Treatment Productivity Increases




Back

# Calibrating the Pareto Tail ($k$)

- **Sample.** Rank firms by revenue, keep 95th–99th pct.
- **Transform.** $F_i$ = empirical CDF $\Rightarrow$ $z_i = -\ln(1 - F_i)$ (linearizes a Pareto tail).
- **QQ-regression.** $\ln x_i = \alpha + \underbrace{\beta}_{=(\sigma-1)/k} z_i + \varepsilon_i.$
- **Result.** $\hat{\beta} = 0.87 \quad \Rightarrow \quad \hat{k} = \dfrac{\sigma - 1}{\hat{\beta}} = \dfrac{3}{0.87} \approx \mathbf{3.5}.$
- **Robustness.** Cut-offs from 95th to 99th give $k \in [3.4,\ 3.6]$ ✓




QQ-plot of $\ln x_i$ (log revenue) against $z_i = -\ln\left(1 - F_i\right)$ for the top 5% of firms

Back

# Analytical Moment Conditions for $\mathcal{B}, \gamma$

- Log mean revenue gap (professional minus family firms): $(\sigma - 1)\log\gamma + \log\left[\frac{\gamma^{-(\sigma-k-1)} - 1 + \delta^{\sigma-k-1}}{\gamma^{k} - 1 + \delta^{-k}} \cdot \frac{1-\delta^{-k}}{1-\delta^{\sigma-k-1}}\right]$
- Share of family-managed firms: $\gamma^{-k}\left(1 - \delta^{-k}\right)$

where $\delta = \left[\frac{B}{f(\gamma^{\sigma-1}-1)}\right]^{1/(\sigma-1)}$

Back

# Productivity and Welfare

- **CES Productivity Index**

$$\mathcal{Z} = \left[ \int_1^{\infty} \widetilde{z}(z)^{\sigma-1} \mu(z)\, dz \right]^{\frac{1}{\sigma-1}}.$$

Here $z$ is baseline productivity, $\widetilde{z}(z)$ is realized productivity, $\mu(z)$ is the normalized density of $z$ among active domestic firms in the differentiated sector, and $\sigma > 1$ is the elasticity of substitution.

- **Consumption welfare**

$$W^C = (1-\alpha)^{1-\alpha}\alpha^{\alpha}\frac{\mathcal{I}}{P^{\alpha}} = (1-\alpha)^{1-\alpha}\alpha^{\alpha}\mathcal{I}\left[\rho\mathcal{Z}\left(\frac{M}{1-m/\alpha}\right)^{\frac{1}{\sigma-1}}\right]^{\alpha}.$$

Here $P$ is the differentiated-goods price index; $\alpha$ its expenditure share; $\mathcal{I}$ fixed nominal income; $M$ active domestic firm mass in that sector; $m$ imports divided by $\mathcal{I}$; and $\rho = (\sigma-1)/\sigma$.

- **Welfare including private benefits**

$$\mathcal{W} = (1-\alpha)^{1-\alpha}\alpha^{\alpha}\frac{\mathcal{I} + \mathcal{B}M_{\text{fam}}}{P^{\alpha}}.$$

Here $\mathcal{B}$ is the per-firm nonpecuniary benefit of family management in money-metric units, and $M_{\text{fam}}$ is the mass of active family-managed domestic firms.

Back